Summer 2026

CYB102
CIC-IDS-2017

GROUP 52

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Monitoring Sources

The monitoring sources provide the telemetry, or the collection of data, needed to analyze the network activity.

- From the CIC-IDS-2017 dataset, we obtained:
  - Raw packet captures (.pcap)
  - Network traffic analysis results (.csv - CICFlowMeter)

  ↳ The CSV files specifically contain the results of analyzing the PCAPs

- Although our playbook recommends Wireshark or tcpdump

  ↳ Instead of analyzing the raw multi GB PCAPs in Wireshark, we decided to look through the already organized CSV files using Splunk

Our analysis will revolve around the hypotheses we developed in part 1 of our project.

Monitoring Sources

Hypothesis:

Attack traffic will be much heavier than normal traffic

1 EX: Tuesday - Brute Force - FTP-Patator (9:20 – 10:20 a.m.)

Source IP	Destination IP	Label
172.16.0.1	192.168.10.50	FTP-Patator
192.168.10.15	192.168.10.50	BENIGN
192.168.10.8	192.168.10.50	BENIGN

count	avg_duration	fwd_pkts
7937	4513156.040065516	43632
116	135091.4224137931	1060
114	135360.6754385965	958

The `fwd_pkts` represents the number of data packets sent from the `Source IP` → `Destination IP`.
It tells you how much actual physical traffic the source pushed toward the victim.
As seen in the photo, compared to the `BENIGN` ones, the number of `fwd_pkts` is significantly bigger.

Not Always True:

The hypothesis mostly stands in the case of high volume based attacks (like FTP-Patator) but not for low rate attacks like slowloris.

EX: Wednesday
DoS slowloris (9:47 – 10:10 a.m.)   DoS Slowhttptest (10:14 – 10:35 a.m.)

Source IP	Destination IP	Label
172.16.0.1	192.168.10.50	DoS GoldenEye
172.16.0.1	192.168.10.50	DoS Hulk
172.16.0.1	192.168.10.50	DoS Slowhttptest
172.16.0.1	192.168.10.50	DoS slowloris
172.16.0.1	192.168.10.51	BENIGN
192.168.10.12	104.105.78.158	BENIGN

count	avg_dur	avg_pkts
10293	23127222.217137862	5.9044010492567764
231073	57081731.892605364	5.279656212538895
5499	57719891.73449718	5.741771231132933
5796	56554365.02122153	6.337129054520359
1	1001027	3
1	115967346	18

As you can see in the highlighted area: slowloris + Slowhttptest spreads packets in a small amount (avg_pkts) over a massive duration (avg_dur).

To find all four, instead of focusing on just attack traffic, focusing on both connection count and avg_dur would be more effective.

Note: GoldenEye + Hulk generate a high packet rate.
FPS = Rate = quantity / time
(Flow per Second)
**count = quantity*   **avg_dur/1000000 = time*

- 226.5 FPS
- 13.2 FPS

- 4.4 FPS
- 4.2 FPS

Monitoring Sources

Hypothesis:

Most attacks will target only a few computers or services.

2 EX: Tuesday - Brute Force (FTP & SSH Patator) + Wednesday - DoS

Source IP	Destination IP	Label
172.16.0.1	192.168.10.50	FTP-Patator
192.168.10.15	192.168.10.50	BENIGN
192.168.10.8	192.168.10.50	BENIGN

Source IP	Destination IP	Label
172.16.0.1	192.168.10.50	DoS GoldenEye
172.16.0.1	192.168.10.50	DoS Hulk
172.16.0.1	192.168.10.50	DoS Slowhttptest
172.16.0.1	192.168.10.50	DoS slowloris

The majority of the attacks on these days targeted `Destination IP` 192.168.10.50 on: port 80 (DoS), 21 (FTP), 22 (SSH)
In the Insiders Network, there were 12 distinct available victims:

- Web Server 16: 192.168.10.50
- Ubuntu Server 12: 192.168.10.51
- Workstations (192.168.10.5 – 192.168.10.25)

The attack traffic overwhelmingly concentrated on just one machine

Most attacks will target only a few computers OR services.

<u>Not Always True:</u>

The hypothesis mostly stands for "<u>few computers</u>" but not for services.

EX: Friday - Afternoon - PortScan

Source IP	Label
172.16.0.1	PortScan

u_targets	u_ports	count
1	1000	158930

<u>Only one target (IP: 192.168.10.50)</u>

```
source="Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv" Label=PortScan
| stats dc(Destination IP) as target by "Destination IP"
```

✓ 158,930 events (before 8/3/26 2:18:10.000 AM) No Event Sampling

Events Patterns Statistics (1) Visualization

20 Per Page Format Preview

Destination IP	target
192.168.10.50	1

Destination ports are commonly used to identify network services.

`u_ports` has a value of 1000

The attacks targeted many services on one machine.

`u_targets` (Computers / Hosts Targeted)
`u_ports` (Services Targeted)

There were 158,930 flows from the attacker (172.16.0.1), which targeted a single victim host (192.168.10.50) across 1000 ports rather than attacking the several other hosts.

Monitoring Sources

Hypothesis:

Attack traffic will happen repeatedly during certain time periods

3

EX: Wednesday - DoS
- slowloris (9:47 – 10:10 a.m.)
- Slowhttptest (10:14 – 10:35 a.m.)
- Hulk (10:43 – 11 a.m.)
- GoldenEye (11:10 – 11:23 a.m.)

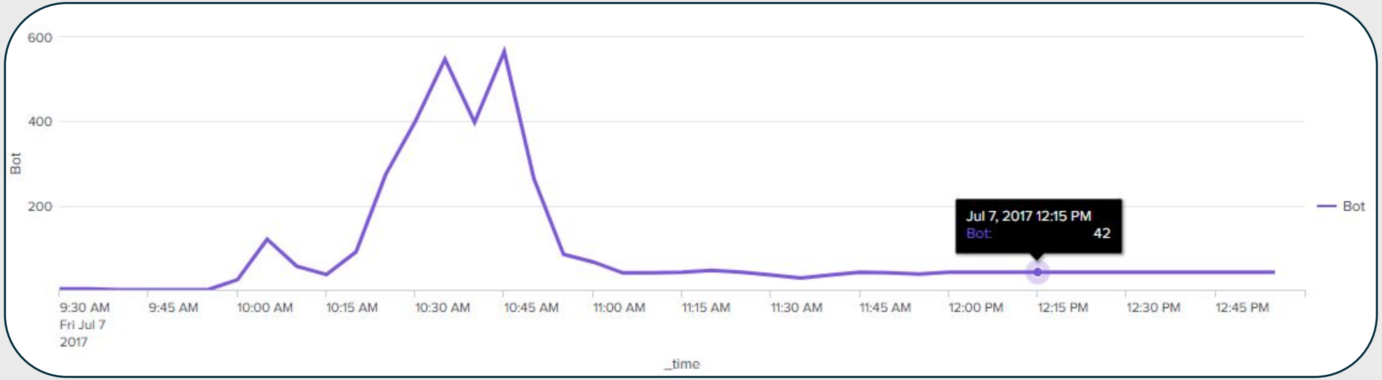
EX: Tuesday - Brute Force
- FTP-Patator (9:20 – 10:20 a.m.)
- SSH-Patator (14:00 – 15:00 p.m.)

As noted on the official source of the dataset, most of the attacks are set to occur in a certain time frame.

Not Always True:

The hypothesis mostly stands for time bound and automated attacks, but not for attacks that rely on stealth.

EX: Friday - Morning - Botnet ARES (10:02 a.m. – 11:02 a.m.)



As seen above, there is a massive spike during the scheduled attack window (10:02-11:02 am), but there continues to be low Bot traffic past even into 12:15 pm.

↳ The graph is consistent with continued command & control (C2) communication, but it isn't sufficient enough evidence to prove that the activity after 11:02 am is actually C2 communication

Playbook - IRM-4-DDOS

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Key performance indicators

This year's objectives	adfasfdsfasfsafa	fdsfsdfasdfs	adfsafasfasfdsafasasdf